Research Session Summary

RLVR Post-Training: Dense Reward, Exploration, Continual Learning, and Multi-Objective Optimization

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Overview

What We Covered Today

Starting from one idea (EGSR), we explored a connected web of research directions spanning four fundamental problems in RLVR post-training:

Problem	Root cause	Ideas explored
Sparse reward	Feedback discarded	EGSR, asymmetric info framework
Mode collapse	Distribution narrowing	Discovery methods (5), population, diverse rewards, multi-objective
Catastrophic forgetting	Sequential training	Continual rehearsal, sleep, self-gen replay, MoD, teacher regularization
Scalar reward	Multi-dim quality compressed	Multi-objective RLVR (4 variants)

Unifying insight: All four problems stem from one root cause — RLVR narrows the policy distribution. Methods that maintain distributional breadth address all four simultaneously.

The Landscape: 15+ Ideas, One Root Cause

RLVR narrows the policy distribution

$\downarrow$ *causes*

Sparse reward

No signal among failures

$\downarrow$

EGSR

Asymmetric info framework

Mode collapse

Diversity destroyed

$\downarrow$

Population RLVR

Multi-objective RLVR

Diverse reward axes

Forgetting

Prior tasks lost

$\downarrow$

$\mathcal{L}_{\text{pred}}$ rehearsal

Sleep consolidation

Mixture of Distillations

EGSR is the foundation: dense reward is prerequisite for multi-objective (needs vector R), continual rehearsal (uses $\mathcal{L}_{\text{pred}}$), population (benefits from smaller G), and diverse rewards (extends to K axes).

Idea 1: EGSR (Primary Project)

EGSR: Self-Distilling Rewards from Environment Feedback

Problem: RLVR uses binary pass/fail, discarding rich execution feedback.

Method: Same model as student (generates) and teacher (judges conditioned on execution feedback). Dense reward $R_{\text{self}} \in [-0.5, 1.0]$ from structured judgment classes.

Maintenance: $\mathcal{L}_{\text{pred}}$ (predict feedback) $+$ $\mathcal{L}_{\text{cal}}$ (anchor to verifier). Label-free, on-distribution.

General framework: Any asymmetric post-hoc information enables self-distilled reward models. Execution feedback is one instantiation. Human rationales, tool responses, checker messages are others.

Key design decisions resolved:

- Reward shaping, not distillation (evaluative not generative advantage)
- "Parameter-aligned," not "on-policy" (precise language)

Idea 2: Discovery vs. Reweighting

Can RLVR Discover, or Does It Only Reweight?

Thesis: RLVR sharpens Pass@1 by collapsing Pass@k. It reweights existing solutions, never discovers new solution strategies.

Prior work check: Heavily explored. Yue et al. (2025) made the diagnosis. POPE (CMU, Jan 2026) solved hindsight relabeling. AEPO, STEER, DIVER, DSDR, ProGRPO address entropy/diversity. DeepSearch embeds MCTS into RLVR. The space is saturated.

Five methods we proposed and their status:

Method	Core idea	Done?	By whom
1. Novelty bonus	Reward distance from mode	Yes	DIVER, AEPO, etc.
2. Hindsight relabeling	Oracle fixes failures, SFT	Yes	POPE (CMU)
3. Latent perturbation	Noise in activations	Mostly	DLR (Jan 2026)
4. Population training	K seeds, cross-pollinate	**No**	—

Idea 3: Latent-Space Exploration

Latent Perturbation for RLVR Exploration

Idea: Explore in continuous activation space instead of discrete token space. Add structured noise to hidden states at decision points during rollout generation. Use verifier reward to credit-assign which perturbations helped.

Key distinction from DLR/Coconut: No architectural changes. Drop-in for standard RLVR. The model still generates tokens — perturbation is a generation-time intervention, not a new reasoning paradigm.

Design ideas: Perturb at high-entropy positions (decision points). Use PCA of solution-pair activation differences for structured perturbation directions. EGSR's self-judge as perturbation evaluator.

Prior work check: DLR (Jan 2026) does latent-space GRPO with an assistant model. Coconut does latent reasoning. Lattice (NeurIPS 2023) does latent perturbation in

Idea 4: Population-Based RLVR

Population RLVR: K Seeds, Cross-Pollinate

Idea: Train K independent RLVR policies (different seeds). Each finds different solution modes. Harvest correct solutions from all, merge into unified student via competence-routed distillation.

Cross-pollination mechanism (from GLM-5):

$$\hat{A}_{i,t} = \sum_k w_k(s_i) \cdot \mathrm{sg}\left[\log \frac{\pi^*_{\theta_k}(y_{i,t} \mid x, y_{i,<t})}{\pi_\theta^{\text{student}}(y_{i,t} \mid x, y_{i,<t})}\right]$$

Teacher-based advantage from each specialist. $G = 1$ suffices (massive throughput gain). Competence-weighted: each teacher only guides problems it solves well.

GLM-5 connection: GLM-5 does this temporally (sequential stages as teachers). We propose it in parallel (independent seeds as teachers). Parallel yields more mode diversity because policies never share training trajectory.

Idea 5: Diverse Reward Signals

Multi-Axis Self-Judgment for Exploration

Idea: Instead of one EGSR teacher prompt, use K prompts evaluating different axes: correctness, novelty, efficiency, robustness, reasoning quality. Disagreement across axes = exploration signal.

$$R_{\text{total}} = R_{\text{self}} + \gamma \cdot \mathrm{Var}_j[R_j(s, a)]$$

Why it helps exploration: Multi-axis reward creates a rugged multi-dimensional fitness landscape. No single solution dominates all axes. The policy is structurally incentivized to maintain diverse strategies (Pareto-optimal on different axis combinations).

Prior work check: RM ensembles exist for robustness (consensus), not exploration (disagreement). Multi-axis grounded self-judgment for RLVR is novel.

Risk: Phase 0 go/no-go — if K different prompts produce correlated axes ($r > 0.8$) at

Idea 6: Multi-Objective RLVR

Beyond Scalar Reward: Multi-Objective RLVR

Observation: The entire RL post-training stack compresses multi-dimensional quality into one scalar. This directly causes mode collapse, objective neglect, and misaligned optimization.

Framework: Maintain vector reward $\mathbf{R} = [R_1, \ldots, R_K]$, optimize over all objectives simultaneously.

Four optimization methods proposed:

Method	Mechanism	Key property
Scalarization	Weighted sum $R = \sum \alpha_k R_k$	Simple but collapses to one point
Pareto GRPO	Keep Pareto-improving rollouts	Principled diversity (**novel**)
Lexicographic GRPO	Priority ordering	Never degrades primary (**novel**)
Conditional	$\alpha(s)$ adapts per problem	Automatic curriculum

Unifies existing methods: Standard RLVR = K=1. Entropy bonus = 2 objectives.

Idea 7: Continual RLVR

Continual Learning for Sequential RLVR

Problem: Sequential RLVR on different tasks forgets prior capabilities. GLM-5 found it severe enough to require a recovery stage.

Key diagnostic (novel): Is RLVR forgetting capability loss or distribution narrowing? Measure Pass@256 at high temperature after sequential RLVR. If it recovers, capabilities are intact — just de-prioritized.

Four solutions proposed:

Method	Training	Mechanism	Novel?
Multi-objective RLVR	Simultaneous	Avoid forgetting by design	Yes
$\mathcal{L}_{\text{pred}}$ rehearsal	Sequential	Predict old-task feedback	Yes
Teacher regularization	Sequential	GLM-5 logit gap on old tasks	Partial
Population / MoD	Parallel	Specialists never train on both	Yes

Three-tier framework: Diagnostic determines intervention. Tier 1 (mild): self-gen

Self-Generating Replay

Idea: During code RLVR, periodically sample math rollouts from current policy, run through math verifier, SFT on correct ones. Replay without a replay buffer.

Key insight: RLVR has a verifier — most CL settings don't. The verifier filters self-generated data for free, always fresh, always on-distribution.

Limitation: Works if and only if the model can still produce correct old-task solutions (Pass@k > 0). The diagnostic directly predicts viability. Fails gracefully at Pass@k ≈ 0.

Prior work: Pseudo-rehearsal and generative replay exist in classical CL. RLEP replays verified solutions within-task. Verifier-filtered cross-task self-gen for RLVR is new but incremental.

Verdict: Strong component of three-tier paper, not standalone.

Sleep-Time Consolidation

Idea: Between RLVR stages, add a dedicated "sleep" phase: self-generate across all prior tasks, verify, train with diversity preservation. No new environment interaction, no competing primary objective.

Why sleep, not concurrent rehearsal: During RLVR, the primary task gradient dominates. During sleep, all tasks are reinforced equally. No task dominates. The model integrates capabilities rather than trading them off.

Consolidation hypothesis: Sleep produces better cross-domain representations than sequential training because the optimizer finds shared circuits that serve all tasks. Testable via hybrid math+code problems after sleep vs. after sequential RLVR.

Prior work: Sleep Replay Consolidation (Nature Comms 2022) exists for small ANNs/SNNs. Not applied to LLMs or RLVR. Concept known, application novel.

Idea 8: Mixture of Distillations

Mixture of Distillations: MoE for RLVR at the Policy Level

Idea: Train K RLVR specialists independently (math, code, science). Merge into unified student via competence-routed teacher-based distillation. Each specialist only teaches problems it's good at.

Key mechanism: Per-token routing from specialist logits. Within a single generation, the student follows the math specialist for setup tokens and the code specialist for implementation tokens. Emergent cross-domain reasoning from specialist composition.

Continual learning: Solved by construction — specialists never train on each other's tasks. When task C arrives, train specialist C, re-distill. Append-only specialist library.

Real-world workflow: Fork base model into K copies, RLVR each on different domain (parallel), distill all into one student with routing. Total wall-clock = one RLVR run.

Prior work: Multi-teacher KD exists. GLM-5 cross-stage distillation exists.

Idea 9: Enterprise and Streaming

Enterprise Continual Learning (Discussion, Not a Paper)

Setting: Deployed RLVR model (e.g., tax advisor). New user trajectories arrive over months. Want to update the model to reflect new data.

Key insight: This is **within-domain distribution shift**, not cross-domain forgetting. Different problem from everything else we discussed. Closer to RLHF than RLVR — user corrections are noisy, delayed reward signals, not verifiable.

Can you do streaming updates? Not really, not yet. GPU memory (inference and training can't coexist), stability (batch size 1 updates are noise), catastrophic interference, serving consistency.

Practical industry solution: RAG for factual updates + periodic LoRA retraining + evaluation gates. Boring but correct.

Summary

Prior Work Verdict Across All Ideas

Idea	Novel?	Crowded?	Pursue?
EGSR / Asymmetric info	Yes (combination)	Low	**Yes (primary)**
Discovery vs reweighting	Diagnosis done	Very high	No
Latent perturbation	Gap shrinking	Medium	No
Population RLVR	Yes (parallel topology)	Low	Backburner
Diverse reward axes	Yes (multi-axis self-judge)	Low	Backburner
Multi-objective RLVR	Yes (Lexicographic GRPO)	Low	**Strong follow-up**
Continual $\mathcal{L}_{\text{pred}}$ rehearsal	Yes (RLVR-specific)	Low	**EGSR extension**
Self-generating replay	Incremental	Medium	Component
Sleep consolidation	Novel for LLMs	Low	Component
Mixture of Distillations	Partial	Low	Backburner
Enterprise streaming	Not research	N/A	No

Ranking: NeurIPS-Worthiness

Rank	Idea	Why	Risk
1	**EGSR**	Novel mechanism + theory + clean empirical plan	E1 go/no-go; results must be strong
2	Multi-objective RLVR (Lexicographic GRPO)	New optimization formulations with provable properties	Needs infra that doesn't exist yet
3	Continual RLVR ($\mathcal{L}_{\text{pred}}$ rehearsal)	Clean follow-up to EGSR; diagnostic alone is novel	Diagnostic might show forgetting is mild
4	Mixture of Distillations	Practical, clean story	May be "just engineering"
5	Population RLVR	Least crowded exploration idea; GLM-5 validates merge	Seeds might find the same mode

Deliverables Produced Today

Papers (3):

- Full NeurIPS scaffold (v2, 15pp)
- Mini class project (v2, 5pp)
- Research proposal (10pp)

Slide decks (12):

- EGSR proposal slides (v2)
- EGSR paper slides (v2)
- Asymmetric info framing
- EGSR complete summary
- Discovery vs reweighting
- Latent perturbation

Slide decks (continued):

- Population RLVR (updated w/ GLM-5)
- Diverse reward signals
- Multi-objective RLVR
- Continual RLVR (broad)
- Continual rehearsal (focused)
- Self-gen replay + Sleep consolidation
- Mixture of Distillations
- This summary

Infrastructure:

- Claude Project setup
- All source files (.tex, .qmd, .sty)

The Priority Roadmap

Now (weeks 1–13):

1. E0: Infrastructure validation (verl + vLLM)
2. E1: Signal validation — **go/no-go gate**
3. E2–E6: Full EGSR experiments
4. Write and submit NeurIPS 2026

Cheap additions during EGSR:

5. Continual learning diagnostic (math Pass@k after code RLVR)
6. $\mathcal{L}_{\text{pred}}$ on math during code RLVR (one extra condition)

After EGSR ships: